

# HEVAL

## Human Evaluation of Machine Translation System

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### Project Description

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HEVAL is a web-based Human Evaluation Platform designed to assess and rank Machine Translation (MT) systems for the English–Hindi language pair. While automatic evaluation metrics such as BLEU, METEOR, and NIST provide quantitative scores, they often fail to capture linguistic nuances, semantic accuracy, fluency, and contextual correctness.

HEVAL addresses this challenge by enabling human evaluators to assess translations based on predefined linguistic criteria and generate a comprehensive ranking of machine translation engines. The platform supports multiple user roles including Administrators, Expert Evaluators, and Non-Expert Evaluators. Administrators manage translations, evaluation criteria, and system weights, while evaluators review machine-generated translations and score them against multiple quality parameters. The collected evaluations are aggregated to generate comparative rankings and performance insights for different translation systems. The system serves as a valuable research tool for Natural Language Processing (NLP), Machine Translation evaluation, and linguistic quality assessment.

### Problem Statement

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Machine Translation systems often produce translations that are grammatically correct but fail to preserve meaning, context, fluency, or linguistic accuracy. Existing automatic evaluation metrics cannot fully measure translation quality from a human perspective. Therefore, there is a need for a standardized Human Evaluation Framework that allows experts and non-experts to assess translations based on multiple linguistic criteria and generate reliable rankings for Machine Translation systems.

### Key Features

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#### User Management

- Secure Login & Registration
- Role-based Access Control
- Expert and Non-Expert Evaluators

#### Translation Evaluation

- Evaluate multiple MT outputs
- Score translations using 11 evaluation criteria

- Ranking scale from Ideal (4) to Not Acceptable (0)

## Administrative Controls

- Manage users
- Configure evaluation weights
- View evaluation analytics
- Generate MT engine rankings

## Reporting

- Sentence-wise results
- Evaluator-wise results
- MT Engine performance rankings

## My Responsibilities

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### Frontend Development

- Developed responsive UI using React.js
- Created Login, Registration, Evaluation, and Results dashboards
- Implemented role-based navigation and protected routes

### Backend Development

- Designed REST APIs using Node.js and Express.js
- Implemented JWT Authentication
- Built evaluation and ranking modules

### Database Design

- Designed MongoDB collections
- Managed translation, evaluation, ranking, and user data

### System Integration

- Connected frontend with backend APIs
- Implemented file upload and XML processing workflows
- Generated ranking reports based on evaluator feedback

## Tech Stack

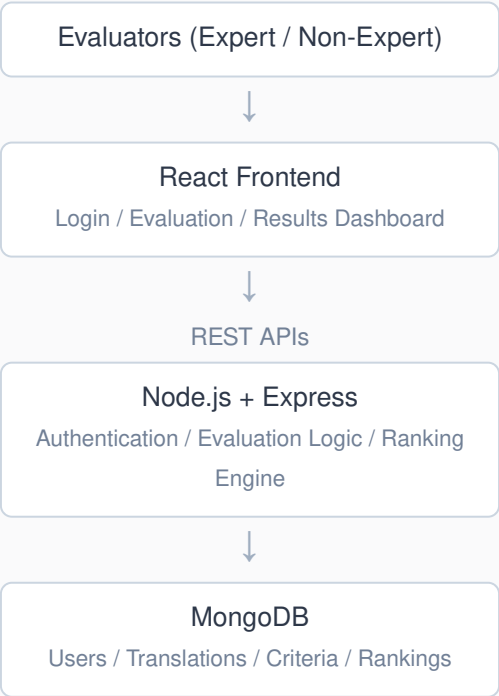
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| LAYER    | TECHNOLOGY                         |
|----------|------------------------------------|
| Frontend | React.js, JavaScript, Tailwind CSS |
| Backend  | Node.js, Express.js                |

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| LAYER             | TECHNOLOGY                           |
|-------------------|--------------------------------------|
| Database          | MongoDB, Mongoose                    |
| Authentication    | JWT                                  |
| File Processing   | Multer, XML2JS                       |
| API Communication | REST APIs                            |
| Deployment        | Web-based Client-Server Architecture |

## System Architecture



## Evaluation Workflow Flowchart

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## Evaluation Criteria

The system evaluates translations meticulously utilizing the following 11 linguistic components:

|                             |                         |
|-----------------------------|-------------------------|
| Gender & Number Translation | Tense Translation       |
| Voice Translation           | Proper Noun Recognition |
| Adjective & Adverb Usage    | Word/Synonym Selection  |
| Sentence Structure          | Punctuation Usage       |
| Emphasis Preservation       | Semantic Accuracy       |
| Overall Translation Quality |                         |